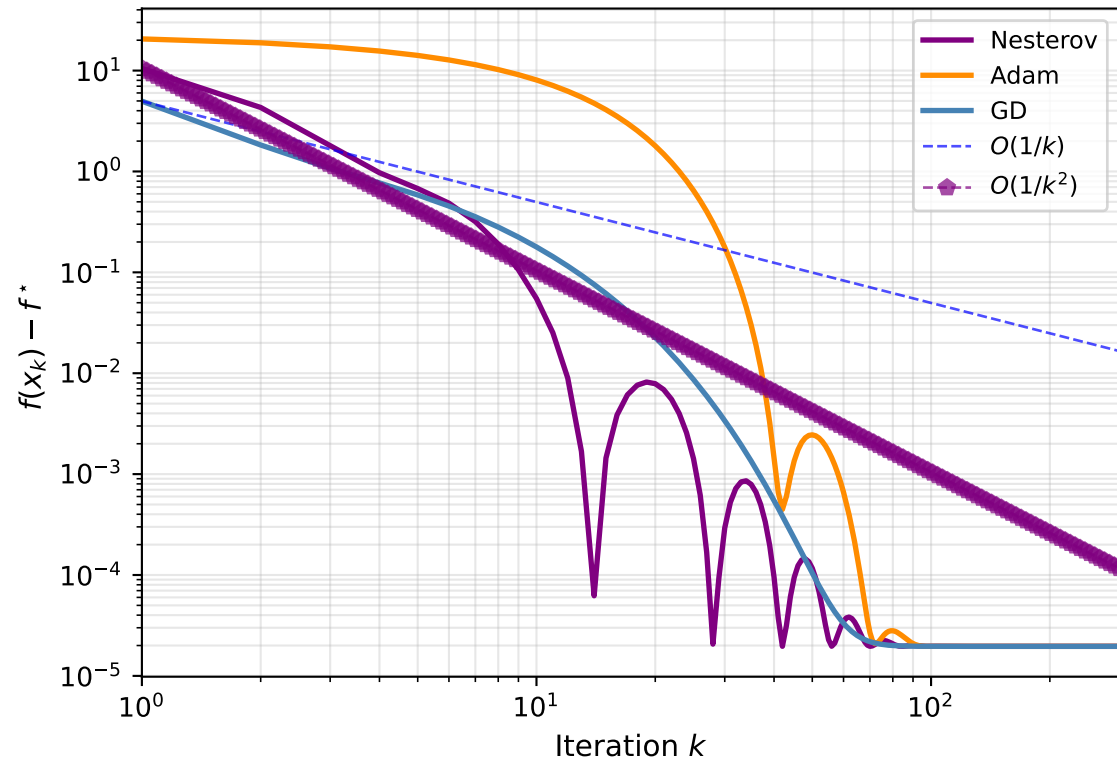


## Why Adam Doesn't Achieve $O(1/k^2)$ : Adaptive Scaling vs Momentum

Convergence Rate: Nesterov vs Adam vs GD  
(Benchmark B, log--log)



Effective Step Size: Nesterov vs Adam  
(Benchmark B)

